

6 Fig. 6.1 is an outline diagram showing two stages of aerobic respiration.

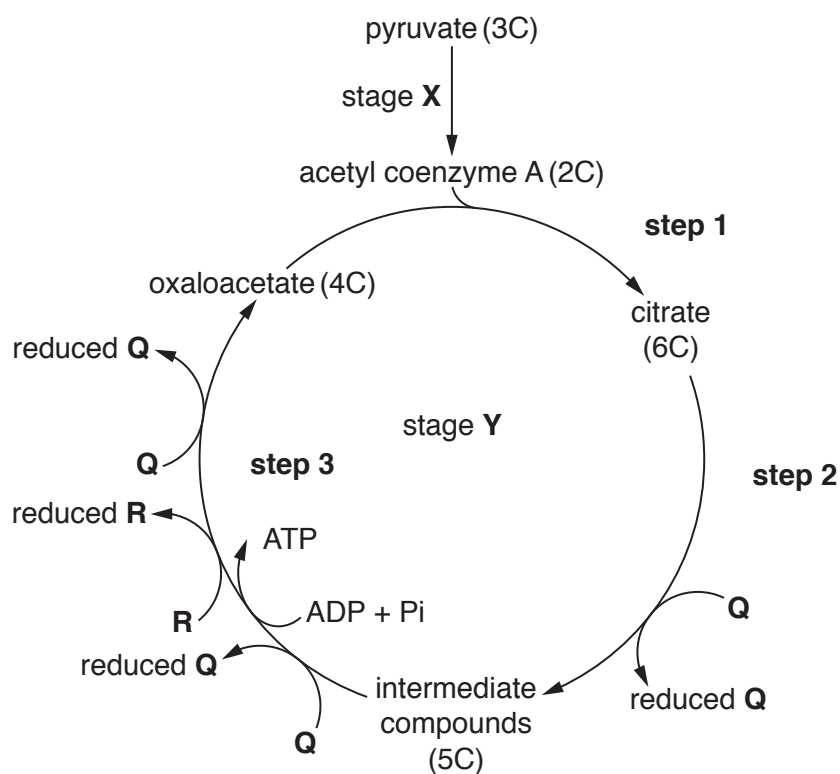


Fig. 6.1

(a) With reference to Fig. 6.1:

name stage **X**

.....

state the precise location of stage **Y**

.....

state the numbered steps in stage **Y** in which decarboxylation occurs

.....

name the type of reaction by which ATP is made during **step 3**

.....

name substances **Q** and **R**.

.....[5]

- (b) Uncontrolled cell division can lead to a cancerous tumour. Many cancer cells break down the amino acid glutamine and convert it to a 5-carbon intermediate compound, which is shown in Fig. 6.1.

Suggest how the breakdown of glutamine can lead to the production of ATP in a cancer cell, other than that directly produced during **step 3**.

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.....[2]

[Total: 7]